IKT200-Assignment 5- Solution

Task 1: Prototyping embedded devices

Answer the following questions. to today's textbook theory: (Chap. 5)

a) What are sensors? What are actuators? Also give 3 examples of each. **Sensor:** a component that receives a signal or one stimulation and responds with an electrical signal. Examples may include microphones, light sensors or rangefinders. **Actuator:** Is a component that converts electrical current or voltage into mechanical movements. Examples may include Lys d iodinated, motors or loud t SPEAKERS.

b) Explain what an Analogue-to-Digital converter does. How high digital resolution does a 10-bit ADC have? How to know this? Analogue-to-Digital converter (ADC) converts an analog ("infinite") electrical signal into a digital (binary) signal. If you want, for example. to load a voltage from a power source, you can use an ADC. With 10 bits, an ADC has a resolution of 1024, ie values from 0 to 1023 (according to the decimal number system , from 0000000000 to 1111111111 binary). A simple way to find out the resolution one on is to take 2^n where n is the number of bits (giving 2^{10} = 1024 in our case) .

c) Pulse-width modulation (PWM) can be used to convert digital to analog signals. How does PWM work? PWM means that you turn a signal on and off at a high frequency, and with a ratio of the conditions that gives an average value of the voltage equal to what you are looking for.

d) Explain cards Breadboard, stripboard and PCB cards. Breadboard - Is a board that you can place components in that do not require soldering, etc. The advantages are very fast assembly and easy to reuse, the disadvantages are that components and wires can easily loosen. Stripboard - Reminiscent of breadboard, but here components and wires are soldered. Forms a much more robust solution than breadboard but also makes it more cumbersome to make changes, and lim s is how much the board can be reused. PCB Card - Common solution for mass production of a printed circuit board. The card is specially designed for the components / circuit it will contain.

e) What is Arduino compared to Raspberry Pi compared to? Understanding about PERFORMANCE operating system (if any), etc. **Arduino:** A microcontroller with some additional components finished soldered onto a circuit board even well selected connection points. Arduino Due has 96KB RAM, 512KB Storage and uses bootloader. **Raspberry Pi:** Is a middle ground between a microcontroller and a full-fledged PC. Has 512MB RAM, SD card with 4GB + Storage. (Thus, it performance is thousands of times stronger than the Arduino.) It uses Linux as OS. **PC:** How we know ordinary laptops or desktops. The most powerful solution in this comparison (and most expensive, but we all have a PC that can "be lent" to a project, it costs nothing).

Task 2: Pathfinding

a) What types of path node definitions have we learned about? What are the pros and cons of them? **Grid:** *Benefits :* Easy to autogenerate, fits nicely to man-made environments. *Disadvantages :* Grids are not suitable outdoors, the distance to neighboring e is not equal in all directions . **Hexgrid:** *Advantages :* All neighboring points are equally close, e RGO look t raining out distances. *Cons :* Fits poorly indoors / in buildings. **Custom Points:** *Advantages :* Fewer points to count on (faster pr o sessering),

points can be placed in the "perfect" position. *Disadvantages :* Frequently requires human follow-up / placement, distances vary.

b) Use the online solution found here: qiao.github.io/PathFinding.js/visual . Show the search algorithms we have been through (all except Depth First Search are in the solution on the link) by taking screenshots after you run each of the links above. Try to set up the path so that the unique with each algorithm is displayed. Give a brief description of what we see and how this differentiates the algorithm from each other. *<<< Not solution suggestions here, but see the same problem (with multiple algorithms) on next week's problem set and solution suggestions. >>>*